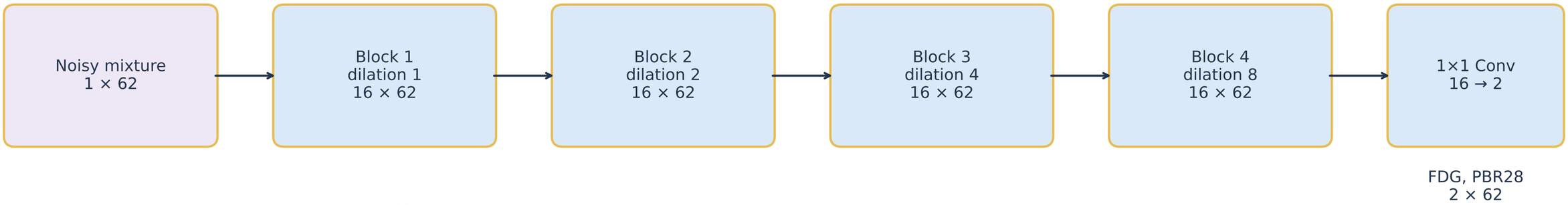
Our TCN: one mixture in, two tracer curves out

Actual prototype architecture • causal sequence-to-sequence separation

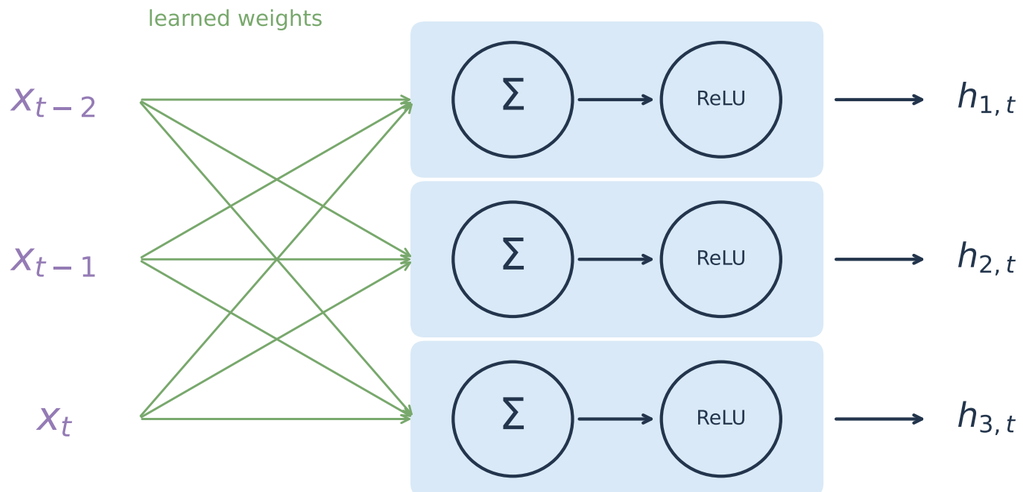
1 THE COMPLETE NETWORK



Shapes are channels × frames for one scan. All 62 output frames are retained.

2 ZOOM IN: ONE CAUSAL CONVOLUTION

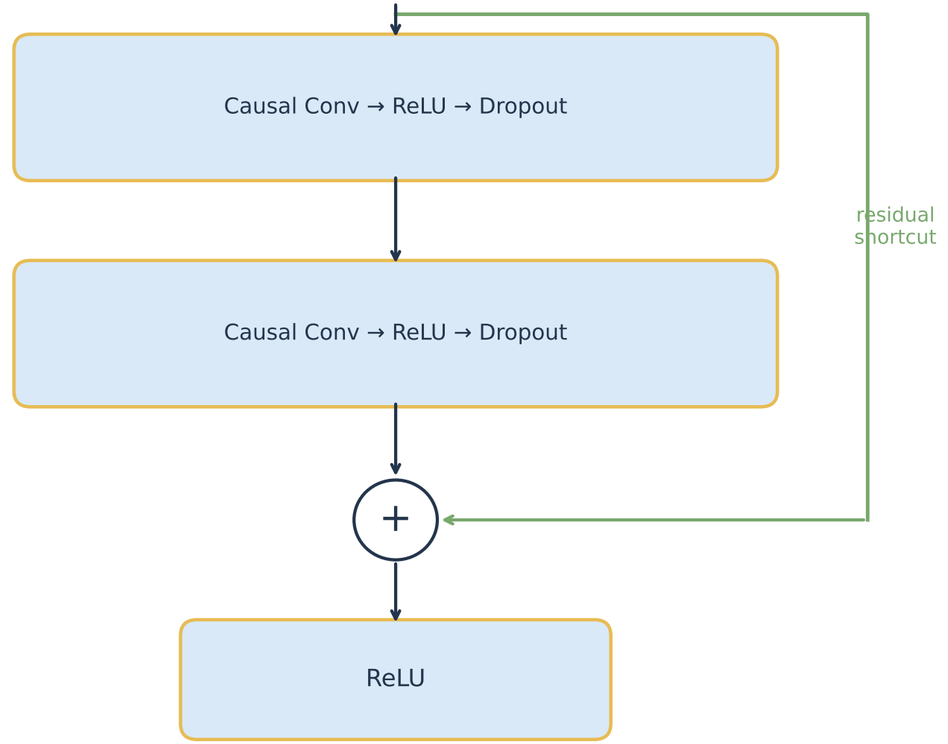
First convolution, kernel size 3, dilation 1



Same filters slide across frames; future inputs are excluded.

Each Σ includes a learned bias. Later convolutions also mix input channels.

3 INSIDE EACH RESIDUAL BLOCK



Block 1 shortcut: 1×1 Conv (1 → 16); others: identity.

Kernel = 3 • Dropout = 0 • Receptive field = 61 frames • Final output is linear (no ReLU)

Inputs are normalized; outputs are physical, decayed TACs. Decay correction occurs afterward. Time spacing varies by frame.